

Saturn: Supplementary Experiments and Analysis

This report supplements the main evaluation of Saturn with benchmark characteristics, a detailed token-cost breakdown, UPO diagnostics, corpus-shift experiments, and heterogeneous-column cases. It uses the same benchmarks and baseline implementations as the main paper, including Pneuma, Solo, and Birdie [1–3].

1 BENCHMARK CHARACTERISTICS

1.1 Query Complexity

Setting. For datasets with SQL annotations, we parse the test queries deterministically and report percentages over the parsable subset. We characterize each query along three axes:

- **Column scope** counts distinct columns referenced by projection and filtering clauses.
- **Table scope** counts ground-truth answer tables; multiple answers are alternative valid tables for retrieval.
- **Reasoning operation** records the dominant intra-table operation, such as filtering, aggregation, or grouping.

FL contains natural-language QA queries and is reported separately from the SQL-based profiles. Dataset abbreviations follow the main paper: Ad (Adventure Works), Ch (ChEMBL), Pb (Public BI), FL/FM (FetaQA), BD (BIRD), Cc (Chicago), and Ow (OpenWikiTable).

Results. Table 1 shows that multi-column queries dominate the SQL-based datasets. Filtering and aggregation are also common. **The benchmarks evaluate table retrieval with substantial multi-column and intra-table reasoning.** Multiple ground-truth tables represent alternative answers; none of the parsable SQL queries contains a JOIN.

1.2 Size-aware Method Behavior

Setting. We order benchmarks by average table size, computed as Avg. Columns $\times$ Avg. Rows, and plot each method’s rank from left to right. Methods that cannot finish because of memory limits are placed at the bottom.

Results. Figure 1 shows that cell-level or context-heavy methods can be competitive on small tables, consistent with observations that many web tables are small [4]. Across mixed table sizes, Saturn remains stable and keeps the most compact representation by storing one bounded UPO-aligned profile per table.

To contextualize the benchmarks, Figure 2 compares them with Hugging Face trending datasets [5]. We collect repositories in the daily trending top-20 between May 1 and June 1, 2025, deduplicate them, and retain those with valid Dataset Viewer size metadata. Their row counts span several orders of magnitude. **This size heterogeneity motivates compact, auditable profiles that remain effective across mixed-size repositories.**

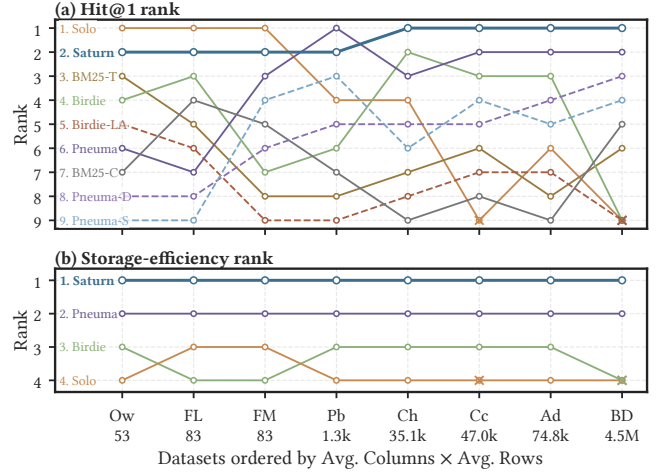


Figure 1: Effectiveness and storage-efficiency ranks, ordered by average table size.

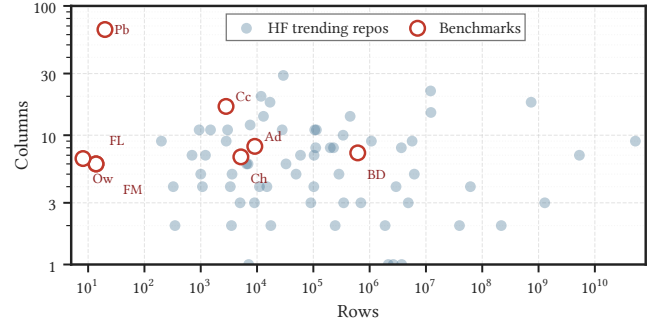


Figure 2: Rows and columns of trending datasets and retrieval benchmarks.

2 TOKEN COST AND SCALABILITY

2.1 Stage-wise Token and Cost Analysis

Setting and accounting. Table 2 expands the main paper’s cost table with conceptualization results for every dataset. Its average is the arithmetic mean across the eight datasets, computed from unrounded token counts. **Offline indexing and online retrieval are reported separately.** The FL indexing total for Saturn includes both population and FL-specific conceptualization; the average conceptualization cost is not substituted into that total.

We use the same model rates as in the main experiment, in US dollars per million input/output tokens: Gemini3 Flash (\$0.50/\$3.00) for conceptualization and Qwen3-Next-80B-A3B (\$0.15/\$1.20) for population and UPO-alignment. Pneuma uses Qwen3-Next for summarization and reranking. Birdie’s tuned Llama-3-8B query generator is estimated with Llama 3 Instruct (8B) rates (\$0.30/\$0.60); its online retrieval generator is mT5-base (0.58B), for which no API dollar estimate is assigned.

Table 1: Query complexity across eight benchmarks. Values are percentages of parsable queries; FL is natural-language QA.

(a) Column and table scope								
Dataset	Queries	Columns			Answer tables			JOIN
		1	2	3+	1	2	3+	
Ad	800	2	31	67	79	10	11	0
Ch	800	1	40	60	95	3	2	0
Pb	800	2	36	61	14	10	76	0
FL	9330	NL-QA			100	0	0	n/a
FM	801	3	33	65	52	14	34	0
BD	1395	10	60	30	92	7	1	0
Cc	802	1	32	66	52	6	42	0
Ow	6602	3	65	32	85	10	5	0

(b) Intra-table reasoning operation						
Dataset	Lookup	Project	Filter	Aggregation	Group	QA
Ad	14	2	43	38	2	–
Ch	17	9	38	34	1	–
Pb	12	5	33	48	1	–
FL	–	–	–	–	–	100
FM	17	0	50	32	0	–
BD	33	1	18	44	4	–
Cc	15	2	42	40	1	–
Ow	68	–	32	–	0	–

Results. Conceptualization requires 448–455 calls and \$0.907–\$1.159 across the eight datasets. On FL, population uses 28,475 calls, 35,501,343 input tokens, and 2,070,890 output tokens (\$7.810). Adding conceptualization’s 453 calls, 1,074,449 input tokens, and 123,271 output tokens (\$0.907) gives the offline total of \$8.717. **Saturn performs more work offline to construct semantically richer profiles, reducing online cost from Pneuma’s \$21.478 to \$9.112.**

2.2 Explicit-prior Scaling

Setting. On FL, we vary the target number of ontology terms $t_c \in \{10, 25, 50, 75, 100\}$, keeping the query budget at 200 and the conceptualization refinement iterations at 5. We measure token consumption in conceptualization (Stage 1) and population (Stage 2), together with transform-cache hit and miss rates.

Results. Figure 3 and Table 3 show conceptualization tokens increasing from 0.55M to 1.07M and population tokens from 26.87M to 68.26M. Each population prompt carries a larger ontology context, but cache hit rate remains above 96%. **Increasing the target ontology size raises prompt cost while preserving substantial transform reuse.**

3 UPO METRICS AND COLD START

3.1 Metrics

Avg. Class Utilization. This metric summarizes LLM judgments of class usefulness for competency questions (CQs). With the CQ set and concept budget fixed, higher utilization indicates greater

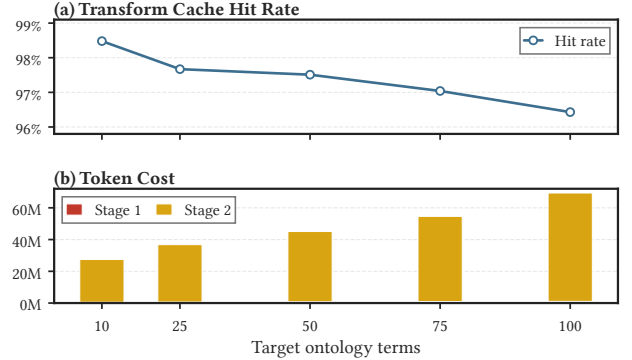


Figure 3: Token overhead and transform reuse as the target ontology size increases.

estimated CQ support per class. Overly coarse classes lose utility when judged insufficient for validation CQs. **We use utilization as a proxy for query fitness, reflecting estimated coverage and compactness.**

Avg. Match Depth. This metric averages the depths of UPO classes assigned to columns during concept grounding. Within a fixed hierarchy, greater depth indicates grounding to more fine-grained concepts. **We use changes in depth to monitor grounding specificity and possible changes in fitness to table data.** A deeper assignment alone does not establish semantic correctness.

Table 2: Token and cost analysis. M denotes million tokens; K denotes thousand calls.

Item	Calls	In (M)	Out (M)	Total (M)	Cost (\$)
UPO conceptualization					
Ad	454	1.14	0.12	1.27	0.939
Ch	451	1.14	0.13	1.28	0.974
Pb	453	1.49	0.14	1.63	1.159
FL	453	1.07	0.12	1.20	0.907
FM	448	1.32	0.13	1.44	1.042
BD	455	1.14	0.13	1.27	0.973
Cc	451	1.15	0.13	1.28	0.959
Ow	453	1.35	0.13	1.48	1.057
Average	452.25	1.23	0.13	1.36	1.001
FL offline indexing					
Pneuma	61.5K	3.36	1.45	4.81	2.246
Birdie	206.6K	2.76*	3.68	6.44	3.037
Saturn	28.9K	36.58	2.19	38.77	8.717
FL online retrieval					
Pneuma	466.5K	128.80	1.80	130.60	21.478
Birdie	9.3K	0.19	0.03	0.22	–
Saturn	9.3K	44.36	2.05	46.41	9.112

* Birdie generates 20 queries per table. The input-token count deduplicates the shared prompt, while calls count individual generations; the dollar estimate follows this accounting convention. Birdie’s additional query-generator and retrieval-generator training costs are excluded. Totals are computed before rounding.

Table 3: Explicit-prior scaling on FL. Token counts are in millions.

t_c	Final classes	Concept. tokens	Population tokens	Total tokens	Cache hit	Cache miss
10	11	0.55	26.87	27.42	98.48%	1.52%
25	29	0.65	36.13	36.78	97.67%	2.33%
50	60	0.82	44.24	45.06	97.51%	2.49%
75	84	0.97	53.64	54.62	97.04%	2.96%
100	105	1.07	68.26	69.33	96.43%	3.57%

3.2 Zero-query Conceptualization

Setting. For each dataset, we synthesize 200 pseudo-queries using Solo’s query synthesis procedure [2], construct the UPO, and populate the corpus. We compare it with a UPO constructed using the original queries. **We evaluate both completed UPOs in a separate post-construction step using the same CQs derived from real user queries within each dataset.** This holds the evaluation requirements fixed while changing the queries used for construction. The main paper separately reports retrieval on the original test queries.

Results. Figure 4 and Table 4 show that average class utilization decreases from 59.0% to 51.6% when synthetic queries replace original queries during construction. **The decrease indicates weaker LLM-judged support for the same real-query requirements.** The trend varies by dataset, with BD and Ow showing higher utilization under synthetic-query construction. Average match depths are numerically close (1.995 vs. 2.005), although the two UPOs have

different hierarchies. Section 4 examines depth changes with each hierarchy held fixed.

Table 4: Cold-start UPO metrics. Utilization changes are in percentage points (pp).

Dataset	Avg. Match Depth			Avg. Class Utilization		
	Original	Synthetic	Δ	Original	Synthetic	Δ (pp)
Ad	2.216	2.250	+0.034	58.2%	47.1%	-11.1
Ch	1.837	1.826	-0.011	53.9%	45.0%	-8.8
Pb	2.051	1.733	-0.318	67.0%	41.3%	-25.7
FL	1.790	2.010	+0.220	70.0%	47.6%	-22.3
FM	2.004	1.951	-0.053	62.4%	52.6%	-9.9
BD	1.748	1.865	+0.117	55.1%	66.8%	+11.7
Cc	2.080	1.922	-0.158	62.0%	51.8%	-10.2
Ow	2.236	2.486	+0.250	43.1%	60.6%	+17.5
Average	1.995	2.005	+0.010	59.0%	51.6%	-7.4

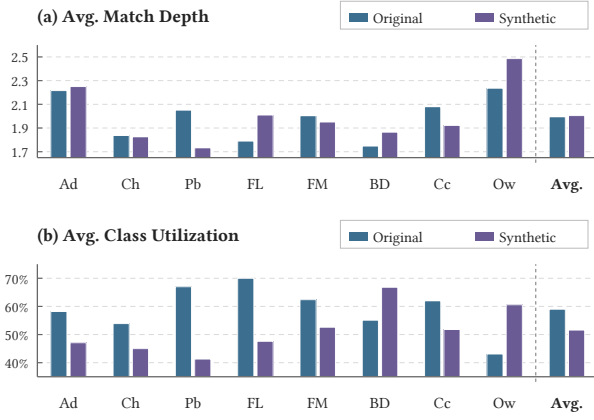


Figure 4: UPO metrics under original-query and synthetic-query conceptualization.

4 CORPUS SHIFT AND UPO EVOLUTION

4.1 Shift Detection

Setting. We use Avg. Match Depth to monitor table-side distribution shifts after UPO construction. We build a UPO on each of Ch, Ad, FL, and FM, then ground columns from every table set with that UPO. Ch and Ad have related coarse table schemas but different fine-grained semantics; FL and FM share tables but have different query sources. The Ch/Ad group and the FL/FM group differ substantially.

Results. In Figure 5, **columns identify the source UPO and rows identify the target table set**. Comparisons within a column hold the UPO fixed. For example, the Ch UPO’s average depth decreases from 1.85 on Ch to 1.49 on FL/FM, while the FL UPO’s depth decreases from 1.79 on FL to 1.16 on Ch. The identical FL/FM rows reflect their shared tables. Depth does not decrease under every transfer, as shown by the FM UPO, so the observed change depends on the source hierarchy. **A drop relative to a fixed UPO’s in-domain depth can signal a need for adaptation.**

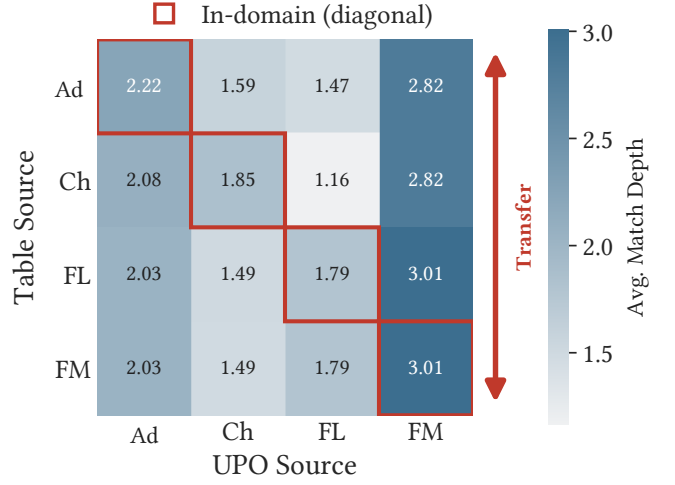


Figure 5: Cross-domain Avg. Match Depth. Each column holds the source UPO fixed.

4.2 UPO Evolution

Setting. After detecting a shift, we reuse the existing UPO as the initial skeleton for conceptualization. Pairwise transfers among FL, Ad, and Ch compare two policies: **Mixed** uses 100 source-domain queries and 100 target-domain queries; **New-Only** uses 200 target-domain queries. We evaluate both UPO metrics on the source and target domains and report their average.

Results. Figure 6 shows that updates emphasizing the target domain improve target-domain UPO metrics while reducing source-domain scores. Average utility remains nearly stable across the two policies. **Mixed offers a better balance between retaining source-domain coverage and adapting to the target.** More advanced continual-learning and concept-pruning strategies are left for future work.

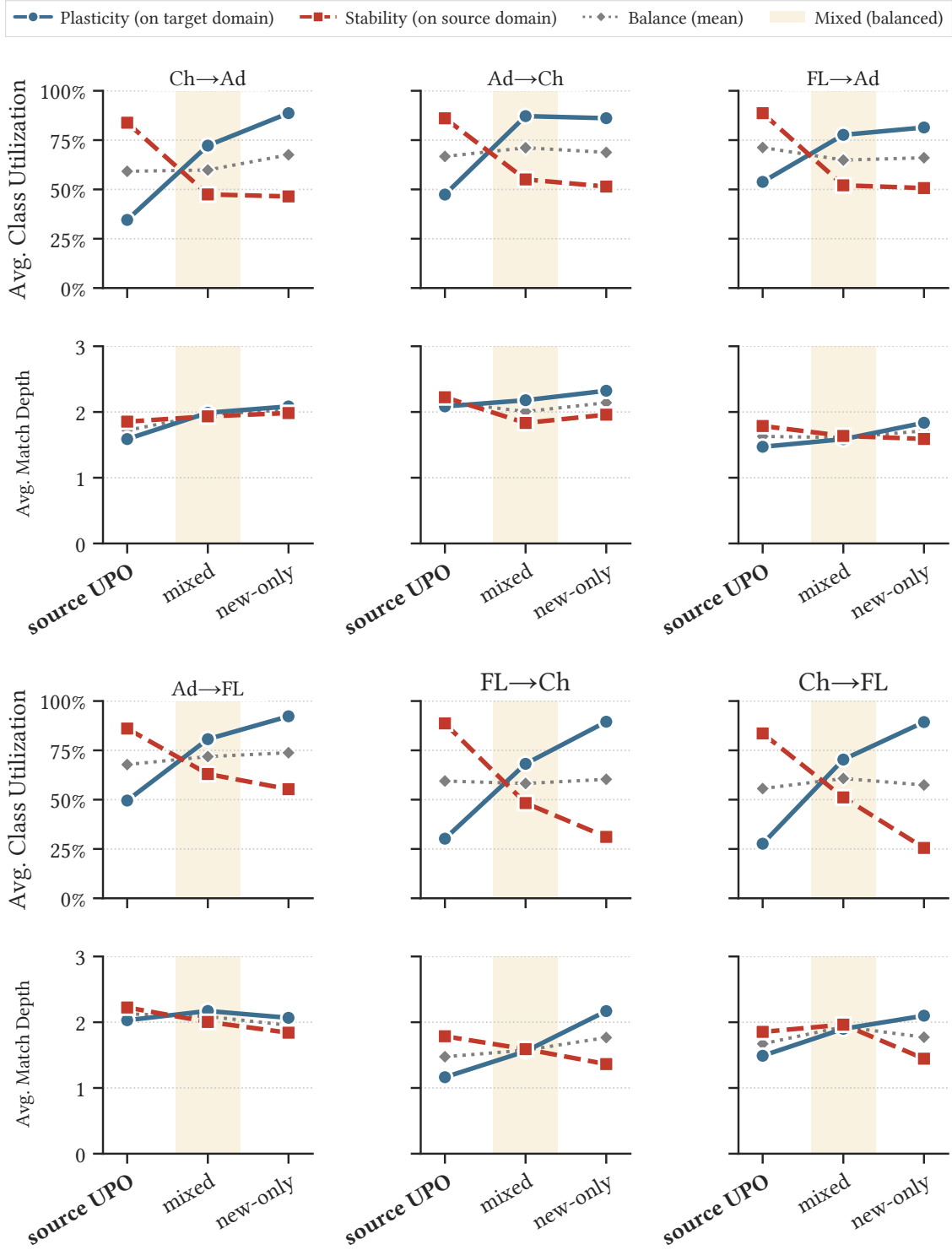


Figure 6: Pairwise UPO evolution under Mixed and New-Only transfer.

5 HETEROGENEOUS-COLUMN TRANSFORM CASES

Setting. We inspect representative high-match transforms from the population stage that were reused by later columns. The cases illustrate the literal formats recognized by generated patterns after sample-based refinement.

Duration formats. A *DurationColumn* pattern covers plain numeric values, percentages, year ranges, compound year-day values, and missing marks. An abbreviated pattern is:

```
^-[0-9]+(\.[0-9]+)?%?$
| [0-9]+--[0-9]+ years?
| [0-9]+ years?, [0-9]+ days?
| -
```

Representative literals include 12, 30%, 4–5 years, 2 years, 73 days, and missing markers.

Historical and approximate dates. A *DateColumn* pattern recognizes historical and approximate forms, including December 1423, 25 November 1400, c. February 1422, and c. 1390s.

Irregular numeric formats. A *NumericColumn* pattern recognizes signed values, decimals, percentages, Unicode minus signs, and dash-like missing markers.

Discussion. These observed patterns demonstrate applicability to heterogeneous literal formats within reusable contracts. Profile synthesis further interprets normalized readouts using the assigned class. More complex settings, including deeply nested JSON fields, heavily mixed-type columns, and highly irregular formats across tables, motivate further study. Future work could combine transform failures and changes in UPO metrics to trigger an LLM-based semantic check of raw samples, readouts, and class definitions in table context. Its feedback could guide transform regeneration or grounding correction. Schema parsing and cell clustering could further support recovery in these settings.

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